Giovanni Luis Gyorody

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Ships software and hardware; seeking a Summer 2027 internship in systems engineering or technical program management

EDUCATION

Embry-Riddle Aeronautical University **Daytona Beach, FL**

Bachelor of Science in Systems Engineering, Enterprise Concentration **Expected December 2028**

Relevant Coursework: Foundations of AI for Engineering, Digital Circuit Design and Lab, Introduction to Systems Engineering, Systems Engineering Design Considerations, Probability and Statistics, Calculus III, Statics, Physics II, Graphical Communications

SKILLS

Systems Engineering: Stakeholder analysis, CONOPS, requirements, system architecture, trade studies, MBSE (SysML, Cameo)
Test and Evaluation: Verification planning, verification and validation, integration and testing, risk management, MOEs and MOPs
AI-Assisted Software: TypeScript, JavaScript, Python, React Native, Expo, Three.js, React Three Fiber, SQL, Supabase, Git, Linux
Engineering Tools: IBM DOORS, CATIA, MATLAB, KiCad, Fusion 360, Ansys, Excel; CFD, structural analysis, PCB design, CNC milling

PROJECTS

Expedition, Squad-Based Habit Platform (iOS) **February 2026 - Present**

- Architected and shipped an iOS app to a 33-user TestFlight beta as its sole technical lead: a React Native and Expo client and a Supabase and PostgreSQL backend with 26 row-level security policies and 11 JWT-verified Edge Functions.
- Led the build as technical lead of a spec-first, AI-assisted workflow; structured audits and on-device verification tracked 85 defects and resolved 64, including 6 security fixes such as an Edge Function trusting a client-supplied user ID.
- Built a custom Three.js and React Three Fiber rendering layer that draws every member's daily trail in one shared squad forest, holding 60 fps on device with 100 trees and an 8-member squad by instancing 400+ environment objects into 3 draw calls.
- Secured the Edge Function layer: JWT verification on all 11 functions with caller identity resolved server-side and never taken from the request body, per-user rate limits of 5 to 20 requests per hour, and idempotency guards on concurrent votes.

Project Initium (Valveless Pulse Jet), Experimental Jet Engine Propulsion Club **September 2024 - Present**

- Owned research and CAD for an 8-person club team, running trade studies across pulse jet configurations that selected a valveless U-shaped design and set the operating point from chamber volume, fuel flow and resonance frequency.
- Modeled the 4-part engine body and its manufacturing drawings in CATIA and ran preliminary Ansys CFD on the design geometry; the body was CNC milled to Embry-Riddle safety requirements, with hot-fire testing pending the team's test stand.

Launch Vehicle Design Project, Team Lead **September 2024 - November 2024**

- Led a team of 4 through the conceptual design of a 3-stage, 134 ft launch vehicle delivering a 5,000 lb payload to LEO, authoring the team charter that defined roles and accountability standards and directing weekly milestones to a 20+ page technical report.
- Sized interstage adapters, fairings, bulkheads and tanks in MATLAB, converging on $23.87M vehicle cost through iterative optimization.
- Presented propulsion selection and cost-benefit trade-offs in a formal design review to peers and faculty.

Glider Design, Analysis and Optimization Competition, 1st Place **November 2024 - December 2024**

- Derived wing geometry, tail configuration, ballast placement and center of gravity from first principles in Excel, then implemented a spin-release launch method used by no competing team, increasing flight distance 178% over a standard throw and taking 1st place.

Motif 65, Mechanical Keyboard Design and Production **May 2024 - August 2024**

- Independently designed and produced a custom PCB keyboard in KiCad with a top-mount architecture, using Fusion 360 FEA to eliminate switch-plate hot spots across 3 prototype builds with an external manufacturer.

EXPERIENCE

Founder and Owner, Egg Hatch LLC **February 2026 - Present**

- Formed and operate the LLC holding Expedition, owning every cross-functional, non-engineering requirement of shipping consumer software independently: entity formation, EIN, operating agreement, business banking and bookkeeping, and annual compliance filings.
- Own release management through the Apple Developer Program and App Store Connect: certificates and provisioning, App Review submissions, App Privacy disclosures, legal and consent flows, and TestFlight beta distribution.
- Instrumented the product with PostHog analytics and Sentry crash observability, and lead go-to-market for a closed beta.

Corporate Relations Chairman, Society of Hispanic Professional Engineers **August 2024 - Present**

- Built and lead a two-director, two-subcommittee organization, Director of Outreach and Director of Events, owning all corporate pipeline development, relationship management and professional programming for the chapter.
- Doubled the chapter's corporate contacts list in one semester through structured warm and cold outreach, and run the chapter's LinkedIn presence, all company-facing communication and post-event recruiter follow-up.
- Organized mixers, resume workshops, mock interviews and info sessions with company representatives, plus a campus-wide event serving 100+ attendees to fund member travel to the SHPE National Convention.

Elementary School Tutor, America Reads America Counts **November 2024 - Present**

- Tutor elementary students in reading and math each academic year under the federal work-study program.